

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1712

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **Sadhana.S (192524298)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

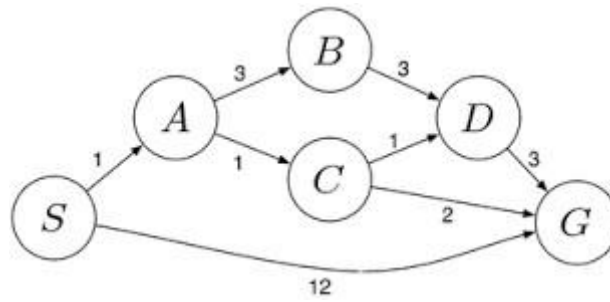
Sadhana.S

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			
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1. Water Jug Problem (4 Gallon & 3 Gallon)

Goal: Obtain exactly **2 gallons** in the 4-gallon jug.

Step	4-Gallon Jug	3-Gallon Jug
Start	0	0
Fill 3-gallon jug	0	3
Pour into 4-gallon jug	3	0
Fill 3-gallon jug again	3	3
Pour into 4-gallon until full	4	2
Empty 4-gallon jug	0	2
Pour remaining 2 gallons into 4-gallon jug	2	0

The 4-gallon jug now contains **exactly 2 gallons**

2.Mars Rover Intelligent Agent

(i) Percepts

- . Camera images
- . Rock and soil samples
- . Temperature readings
- . Atmospheric pressure
- . Terrain information
- . Obstacle detection sensors

(ii) Operating Environment

- . **Observable:** Partially observable
- . **Deterministic:** Stochastic
- . **Sequential:** Yes
- . **Dynamic:** Yes
- . **Continuous:** Yes
- . **Single Agent:** Yes

(iii) Actions

- . Move forward/backward
- . Turn left/right
- . Capture images
- . Collect soil samples
- . Analyze rocks
- . Send data to Earth

(iv) Performance Measure

- Number of successful samples collected
- Accuracy of scientific analysis
- Safe navigation
- Minimum energy consumption
- Successful communication with Earth

(v) Suitable Agent Architecture

Learning Agent

Reason:

- Learns from previous experiences.
- Adapts to unknown Martian terrain.
- Improves navigation and obstacle avoidance.
- Makes better decisions over time.

3. 8-Queens Problem Formulation

Initial State

Empty chessboard.

State Space

All possible arrangements of 8 queens.

Actions

Place one queen in a safe position in each row.

Goal Test

No two queens attack each other:

- Same row
- Same column
- Same diagonal

Path Cost

Each queen placement has equal cost.

Total cost = 8 placements.

Search Strategy

Backtracking Search / Depth First Search
(DFS)

4. OLA Cab Booking Problem

Type of Problem-Solving Agent

Goal-Based Agent

Reason:

The customer has a goal of reaching the destination by selecting the most suitable cab.

Pseudocode

START

Input pickup location

Input destination

Display available cab types

(Mini, Micro, Sedan, Prime, Shared)

Customer selects cab

Calculate fare

Estimate arrival time

If cab available

 Book cab

 Send driver details

Else

 Display "Cab Not Available"

END

5. Uniform Cost Search (UCS)

Given Edges:

- . $S \rightarrow A = 1$
- . $S \rightarrow G = 12$
- . $A \rightarrow B = 3$
- . $A \rightarrow C = 1$
- . $B \rightarrow D = 3$
- . $C \rightarrow D = 1$
- . $C \rightarrow G = 2$
- . $D \rightarrow G = 3$

UCS Steps

Step	Expanded Node	Cost
1	S	0
2	A	1
3	C	2
4	D	3
5	G	4 (Goal Reached)

Optimal Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

$$1 + 1 + 2 = 4$$

Result

Using the **Uniform Cost Search (UCS)** algorithm, the **least-cost path** is $S \rightarrow A \rightarrow C \rightarrow G$ with a **minimum cost of 4**. UCS guarantees the optimal path by always expanding the node with the lowest cumulative cost.